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YASHAS KARANAM
CO-FOUNDER AND COO, BELLATRIX AEROSPACE

first real exposure to space technology as an Isro subcontractor.

"That experience was invaluable," says Azista Space Managing Director Srinivas Reddy. "It taught us the quality systems, documentation discipline and process rigour that space demands, none of which you can learn from commercial electronics work."

For the next few years, Azista Space built hardware to Isro's drawings, to the agency's standards, for its national missions. It had two options: keep building to Isro's specifications and remain a contractor, or venture out on its own. About three years in, it made a call: it would develop its own products for the global market.

Space hardware demands a different kind of discipline. It has to survive an environment that cannot be fully replicated on the ground and not repaired once launched—vacuum, extreme thermal cycling, radiation and severe launch vibration. It has to work the first time, every time.

"That changes everything upstream. We had to build quality systems and traceability where every component, process and test is documented and auditable end to end," says Reddy. "We had to adopt space-grade parts selection and screening, environmental qualification and testing (thermal-vacuum, vibration, EMI/EMC), cleanroom assembly practices, and rigorous failure analysis and review discipline. Reliability and repeatability, not just a working prototype, become the real deliverable."

The policy change in 2020 became the catalyst it needed.

A MIXED INHERITANCE

"The private sector has always been involved. The only difference now, which is a big difference, is that private people can be mission owners," says Ravichandran. "We all worked as contractors. We were building something, but the mission was owned by India."

Isro's vendor base could make almost any part to a specification, but it was thin where the requirement was to design, qualify and repeatedly manufacture a specialised part of its own: aerospace-grade valves, cryogenic components, radiation-hardened electronics, optical glass and coatings. So companies had to build these in-house or import.

Immanuel Louis, Co-founder and COO of Astrophel, maps the gap precisely. "India has strong precision-manufacturing capabilities, but gaps remain in aerospace qualification, traceability and low-volume specialised supply chains," says Louis. For example, most Indian companies still import high-pressure and cryogenic valves often from Germany, so Astrophel now builds its own. Roughly 47% of the company's bill of materials by value still comes from outside India.

The harder lesson, Louis says, came after the engineering. "Industrialising the technology is often harder than developing it," he says. "Aerospace is not only a technology problem in India. It is a manufacturing, supply-chain and execution problem." The machining India could do. The qualified, specialised supply, it largely could not.

That gap forced companies to build capabilities in-house. Propulsion was perhaps the clearest case, because there was almost no domestic manufacturing base. Bellatrix, for example, set out to build engines for a new generation of small satellites and found the technology had to be reinvented for smaller scale.

"Initially, the industry was largely focused on the GEO market. We recognised the shift toward LEO early and had to re-invent the wheel, thereby developing solutions specifically for lower-orbit satellites. This meant inventing entirely new products for the LEO regime," says Karanam. "At the time, none of the existing players had strong, active R&D teams—they were primarily manufacturers." Both the company's customers and suppliers were based overseas, and, as Karanam puts it, "there were absolutely no suppliers in the country".

GEO, or Geostationary Earth Orbit, and LEO, or Low Earth Orbit, are the two main types of orbits used for satellites. LEO is closer to earth and satellites in it move quickly, typically at altitudes of 160 to 2,000 km. GEO is much farther away, at about 35,786 km above the equator, where satellites orbit at the same rate as